EFFECTS AND COEFFECTS ENTWINE ON CONTAINERS: ONE MONAD, TWO FEEDS, AND THE BRANCHING OBSTRUCTION

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ABSTRACT. A single monad M on Set acts on the category Cont of containers along either axis: on *shapes* it gives the Ahman–Bauer effect monad T_M; on *positions* it gives, by contravariance, a coeffect comonad G_M. We ask how these two feeds of one monad interact. They always entwine: for every M with a $\prod$-cointerpretation lifting there is a canonical mixed distributive law $\lambda\colon T_M G_M \Rightarrow G_M T_M$ — the oplax product-comparison of M, read on positions — making G_M a comonad on T_M-algebras and T_M a monad on G_M-coalgebras. This Turi–Plotkin bialgebra holds unconditionally, nondeterminism included. The reverse orientation is another story: the compositor $\kappa\colon G_M T_M \Rightarrow T_M G_M$ that would let one *sequentially compose* effect–coeffect programs $G_M p \to T_M q$ into a category exists if and only if M is *non-branching* (every operation has arity at most one), a class we identify exactly as the writer-with-absorbing-exceptions monads $MX \cong E + A \times X$. There the arrows form a Hughes arrow, equivalently a Freyd category over $(\mathsf{Cont}, \times)$; for branching M they fail, through two independent axioms. Branching obstructs the *orientation* of the effect–coeffect interaction, not the interaction itself.

1. Introduction

A monad is a model of an *effect*: a way of building a computation that returns a value while also doing something — failing, choosing, logging, diverging. A comonad is a model of a *coeffect*: a way of consuming a computation that reads a value together with some ambient *context* — an environment, a history, a resource budget. The two notions are formally dual, and a long line of work in programming semantics has asked how they interact when both are present at once: how does one *compose* a program that reads context and produces effects with another of the same shape? The classical answers — Hughes arrows [6], Freyd categories [12, 2], biKleisli categories of a monad and a comonad with a distributive law [13, 15], and the bialgebraic operational semantics of Turi and Plotkin [14] — all live over an arbitrary

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base category and take the monad and the comonad as *independent* data, related (if at all) by a distributive law that must be supplied by hand.

This paper studies the case where the monad and the comonad are not independent but are the *two feeds of a single monad*, and where the base category is the category Cont of containers (equivalently, polynomial functors in one variable). The setting makes the interaction canonical, and canonicity turns out to have sharp consequences.

The two feeds. A container (S, P) is a set S of *shapes* together with a family $P \colon S \to \mathsf{Set}$ of *positions*; it presents the polynomial endofunctor $[\![S, P]\!]Y = \sum_{s \in S} Y^{P(s)}$ on Set. A morphism of containers is a map *forward* on shapes and *backward* on positions — positions are contravariant — and this single fact about variance is the hinge of the whole paper. Given any monad M on Set (with a mild support hypothesis, made precise in §2), there are exactly two ways to feed M into a container, and the variance decides the outcome:

- *Along the shapes*, $T_M(S, P) = (MS, P^\star)$ applies M to the shape set and extends the positions to match (Ahman–Bauer [1], Thm. 6.3). Shapes are covariant, so T_M is a **monad** on Cont — the *effect* feed.
- *Along the positions*, $G_M(S, P) = (S, M \circ P)$ applies M to each position fibre and leaves the shapes fixed (the *transfer*). Positions are contravariant, so G_M is a **comonad** on Cont — the *coeffect* feed.

There is no third axis and no freedom in the outcome. The reader should hold onto the asymmetry: it is the same variance — positions run backward — that makes G_M a comonad and, we will find, forces the single direction in which the effect feed can pass through the coeffect feed.

The question, and the answer. Do the two feeds T_M and G_M of one monad M interact via a distributive law, and if so, what does that law model? We give a complete answer, and it is a *dichotomy of two dual laws* that coincide only in a degenerate regime.

Theorem 1.1 (Main dichotomy)**.** *Let M be a Set-monad with a $\prod$-cointerpretation predicate lifting (for instance Pf, Maybe, Writer, or* Id*), with effect monad T_M and coeffect comonad G_M on Cont as above. Then:*

(1) (Bialgebra face, all M.) *There is a canonical mixed distributive law $\lambda \colon T_M G_M \Rightarrow G_M T_M$, the oplax product-comparison of M read on positions, satisfying the four entwining axioms for* every *such M. Consequently G_M lifts to a comonad on T_M-algebras and T_M lifts to a monad on G_M-coalgebras: a Turi–Plotkin-style λ-bialgebra.*

(2) (Arrow face, non-branching M only.) *The reverse compositor $\kappa \colon G_M T_M \Rightarrow T_M G_M$ satisfies the four axioms — equivalently, the effect–coeffect programs $p \rightsquigarrow q := \mathsf{Cont}(G_M p, T_M q)$ form a category, indeed a Hughes arrow / Freyd category over* $(\mathsf{Cont}, \times)$ — if and only if M is non-branching, *meaning every shape of M has arity at most one. The sole obstructed axiom is associativity.*

(3) (Classification of the positive class.) *A cartesian M is non-branching if and only if it is a* writer-with-absorbing-exceptions *monad*

$$MX \;\cong\; E + A \times X, \qquad (A, \cdot, e) \text{ a monoid}, \quad (E, \odot) \text{ a left } A\text{-set},$$

with writer unit $x \mapsto (e, x)$ and the log A acting on the exceptions E; the arrow category exists throughout this class.

Parts (1)–(3) are Theorems 4.1, 6.1–6.6, and 7.1 below. The single monad M carries two comparison maps between its composites $T_M G_M$ and $G_M T_M$, and they behave oppositely: the oplax one runs $T_M G_M \Rightarrow G_M T_M$ and is always a law; the lax one runs the other way and is a law only when M does not branch. *Same monad, same two feeds, same category* Cont*: the unification of effects and coeffects exists unconditionally as a bialgebra, and its categorical packaging as an arrow calculus is exactly what branching obstructs.* The obstruction is not to the interaction but to its *orientation*.

Why this is worth stating. Three points give the result its edge.

First, the bialgebra face *keeps the branching monads*. A recurring worry about effect–coeffect calculi is that the interesting computational monads — nondeterminism Pf, probability, lists — are precisely the ones that break the sequential composition. They do break the *arrow*, but they do not break the *bialgebra*: part (1) holds for Pf and List exactly as for Maybe. So the denotational, bialgebraic unification of effects and coeffects on containers is available for every M, and the affirmative answer to "is there a Turi–Plotkin bialgebra here?" is unconditional.

Second, the obstruction is *named*. Part (3) does not merely say "non-branching"; it identifies the class as the writer-with-absorbing-exceptions monads $E + A \times (-)$, a recognisable and closed family (the writer transformer over the exception monad, twisted by an action). "Non-branching" is moreover a genuinely new dividing line, provably orthogonal to commutativity and affineness (Proposition 7.5): Pf is commutative and branching, $E + (-)$ with $|E| > 1$ is non-commutative and non-branching, the distribution monad is affine and branching.

Third, branching disables the arrow *twice over*, through two independent axioms. The compositor κ fails associativity (Theorem 6.1) through the multiplication of M *merging* leaves (a *union of products versus product of unions* discrepancy); and, separately, the effect monad T_M has no *natural* strength for the product on Cont (Lemma 6.4), failing through leaf-*symmetry* rather than merging. Both failures are equivalent to the arity condition, by different arguments — a pleasingly redundant obstruction.

Method. Because both feeds are identity on shapes over MS (for T_M) or on S (for G_M), every law and every axiom localises to a *position-level equation in* Set, with backward maps composing contravariantly. At a shape with leaves b and position sets Z_b, the two composites $T_M G_M$ and $G_M T_M$ read as $M(\prod_b Z_b)$ and $\prod_b M(Z_b)$, and the only canonical map between them is the oplax product-comparison $\mathrm{st}\colon M(\prod_b Z_b) \to \prod_b M(Z_b)$. The entwining axioms are then naturality squares for M's unit and multiplication; the arrow's obstruction is that the *reverse* (lax) comparison fails to commute with the leaf-merging in μ^M; and the strength obstruction is a Yoneda computation showing a natural distributivity map must read a single distinguished leaf, which leaf-symmetry then forbids. The classification is a polynomial-functor argument (arity ≤ 1 is degree ≤ 1) followed by a monad-structure pin.

Outline. Section 2 fixes containers, the two liftings, and the entwining vocabulary. Section 3 presents the effect monad T_M and the coeffect comonad G_M and the sense in which they are one monad's two feeds. Section 4 proves the bialgebra face — the law λ for all M — and Section 5 shows the reverse orientation fails, isolating branching as the obstruction. Section 6 builds the arrow face: the biKleisli category (Theorem 6.1), the Freyd/Hughes structure and the strength obstruction (Theorem 6.6). Section 7 classifies the positive class as writer-with-absorbing-exceptions and closes the associativity axiom across it (Theorem 7.1). Section 8 reports the machine-checked instance. Section 9 places the result among its neighbours, and Section 10 concludes.

Provenance. The mathematical results collected here were established in a sequence of the author's proof notes; the effect monad T_M is due to Ahman and Bauer [1], and the entwining and arrow formalism is classical (Beck [4]; Power–Watanabe [13]; Brzeziński–Majid [5]; Uustalu–Vene [15]). What is new is the identification of the law welding one monad's two container feeds, its forced orientation, the branching obstruction, and the classification of the positive class. Attribution is given in place. Two axioms are proved in coordinates and machine-verified on the flagship monads but not written out as a fully symbolic chase over an arbitrary predicate lifting; these mechanical gaps are flagged honestly where they occur (Remarks 4.3 and 7.3).

2. Preliminaries

We work with containers and with Set-monads carrying a support structure. This section fixes notation and recalls the entwining vocabulary; a reader fluent in polynomial functors and distributive laws can skim it.

2.1. **Containers.** A *container* is a pair (S, P) with $S \in \mathsf{Set}$ (*shapes*) and $P \colon S \to \mathsf{Set}$ (*positions*). A *morphism* $(S, P) \to (T, Q)$ is a pair (u, f) with $u \colon S \to T$ *forward* and, for each $s \in S$, a map $f_s \colon Q(u\, s) \to P(s)$ *backward*; composition is forward on shapes and backward (in the opposite order) on positions. Containers and their morphisms form a category Cont. The *extension* functor

$$[\![-]\!] \colon \mathsf{Cont} \longrightarrow [\mathsf{Set}, \mathsf{Set}], \qquad [\![S, P]\!]Y = \sum_{s \in S} Y^{P(s)},$$

is fully faithful, with image the *polynomial* endofunctors; we freely identify a container with the functor it presents. Finite products in Cont are

$$(S, P) \times (T, Q) = \big(S \times T,\ (s, t) \mapsto P(s) \sqcup Q(t)\big),$$

matching $[\![p \times c]\!]Y = [\![p]\!]Y \times [\![c]\!]Y$; this is the cartesian tensor used in §6.

2.2. Set-monads with support and the predicate lifting.

Let $M = (M, \eta, \mu)$ be a monad on Set presented as a polynomial functor

$$MX \;=\; \sum_{s \in M1} X^{\,\mathrm{ar}(s)}, \qquad \mathrm{ar}\colon M1 \to \mathsf{Set},$$

so each element $m \in MX$ has a *support* of *leaves* $b \in \mathrm{lv}(m)$ carrying labels $x_b \in X$. We say M is *non-branching* if $|\mathrm{ar}(s)| \leq 1$ for every shape s (equivalently every m has $|\mathrm{lv}(m)| \leq 1$), and *branching* otherwise. Flagship examples: $\mathsf{Maybe} = 1 + (-)$, exception $E + (-)$, $\mathsf{Writer}_N = N \times (-)$, and Id are non-branching; finite powerset Pf and List are branching.

To lift M into positions we need a *predicate lifting* $(-)^\star$ turning a family $P\colon S \to \mathsf{Set}$ into $P^\star\colon MS \to \mathsf{Set}$. We use the $\prod$-*cointerpretation* of Ahman and Bauer [1, §6.1,§6.5]:

$$P^\star(m) \;=\; \prod_{b \in \mathrm{lv}(m)} P(x_b),$$

the universal (product-based) lifting. Its unit datum is the projection i out of the singleton product $P^\star(\eta_S s) = \prod_{\{*\}} Ps = Ps$, and its multiplication datum is the restriction j that reindexes a product along the leaf-map of μ^M. The monads for which this weak Mendler algebra is defined include all the flagship examples above; the reader monad A^K is *not* one, which is why the lifting is taken weak. We call such M a $\prod$-*cointerpretation monad*; this is the standing hypothesis throughout, and *cartesian* means μ^M (and η^M) are cartesian natural transformations (preserve the leaf-count).

Remark 2.1 ("Non-branching" is not Kock-affine). Non-branching means *arity* ≤ 1, i.e. degree ≤ 1 as a polynomial (a constant term plus a linear term). It is *not* affineness in Kock's sense $M1 \cong 1$: here the shape set $M1$ may be arbitrarily large. The two coincide only trivially ($M1 \cong 1$ forces $M = \mathrm{Id}$). We say *non-branching* or *arity*-≤ 1 throughout and never *affine*, to avoid the clash.

2.3. Mixed distributive laws and entwinings.

Let T be a monad and G a comonad on a category $\mathcal{C}$. A *mixed distributive law* (or *entwining*) of G over T is a natural transformation $\lambda\colon TG \Rightarrow GT$ satisfying four axioms — two expressing compatibility with the unit and multiplication of T, two with the counit and comultiplication of G (Beck [4]; Power–Watanabe [13]; the entwining-structure language is Brzeziński–Majid [5]). Such a λ is exactly the datum lifting G to a comonad on the category $\mathrm{Alg}(T)$ of T-algebras and T to a monad on the category $\mathrm{Coalg}(G)$ of G-coalgebras; a compatible T-algebra-and-G-coalgebra is a λ-*bialgebra* in the sense of Turi and Plotkin

[14]. We write the four axioms E1–E4 (unit-T, mult-T, counit-G, comult-G) and, for the opposite orientation $\kappa\colon GT \Rightarrow TG$, their mirror E1′–E4′; here E2′ is the mult-T/associativity axiom.

Dually, a mixed distributive law $\kappa\colon GT \Rightarrow TG$ lifts G to a comonad on the Kleisli category $\mathrm{Kl}(T)$, whose *coKleisli* category is the *biKleisli* category of the pair: its objects are those of $\mathcal{C}$, its morphisms $p \to q$ are the maps $Gp \to Tq$, and composition uses κ to thread the effect of one map past the coeffect of the next [13, 15]. When T is strong and G costrong compatibly with κ, this biKleisli category is a *Hughes arrow* [6], equivalently a *Freyd category* over the cartesian base [12, 2, 8]. These are the two structures the paper produces from the two feeds of one M.

3. The two feeds of one monad

We recall the two liftings and the exact sense in which they are dual halves of one monad.

3.1. **The effect monad.** Ahman and Bauer [1, Thm. 6.3] show that a $\prod$-cointerpretation monad M lifts along shapes to a monad on Cont:

$$T_M(S,P) = (MS, P^\star), \qquad P^\star(m) = \prod_{b\in \mathrm{lv}(m)} P(x_b).$$

The unit η^T covers η^M on shapes and is backward the singleton projection i; the multiplication μ^T covers μ^M and is backward the restriction j. Because shapes are covariant, T_M is a monad. On extensions $[\![T_M(S,P)]\!]Y = \sum_{m\in MS} Y^{P^\star(m)}$: T_M is the container of *M-structured computations over the operations of* (S,P), with positions the tuples of operands, one per leaf.

3.2. **The coeffect comonad.** Dually, apply M to the position fibres:

$$G_M(S,P) = (S, M\circ P).$$

The counit ε is backward η^M and the comultiplication δ is backward μ^M; shapes are untouched. Because positions are *contravariant*, the covariant monad M becomes a *comonad* G_M on Cont. On extensions $[\![G_M(S,P)]\!]Y = \sum_s Y^{M(Ps)}$: an operation of arity $P(s)$ acquires an M-structured arity, a *context* wrapped around every operand slot. (For $M = \mathsf{Maybe}$, G_M adjoins a fresh basepoint to each fibre — a default or exception argument on every operation.)

The two liftings are welded together by the fibrewise $(-)^{\mathrm{op}}$ that makes a container a container: apply M where the variance runs forward and monads stay monads; apply it where the variance runs backward and monads become comonads. There is no third axis, and no choice in the outcome. The rest of the paper studies how these two feeds of *the same* M interact.

4. The bialgebra face: the two feeds always entwine

This is the headline. We show the two feeds interact via a canonical mixed distributive law $\lambda\colon T_M G_M \Rightarrow G_M T_M$, valid for *every* $\prod$-cointerpretation M, branching or not, and identify what it models.

4.1. The two composites and the canonical comparison. Both $T_M G_M$ and $G_M T_M$ are the identity on shapes over MS, so it suffices to read their positions at a fixed shape $m \in MS$. Localise at m, write $Z_b := P(x_b)$ for the position set at leaf $b \in \mathrm{lv}(m)$, and compute:

functor	positions at m
$G_M\, T_M$	$M(P^\star m) = M(\prod_b Z_b)$
$T_M\, G_M$	$(M \circ P)^\star(m) = \prod_b M(Z_b)$

($G_M T_M$: run T_M, then fatten its positions with M. $T_M G_M$: fatten first, then take the product-lifting of the M-fattened positions.) Between $M(\prod_b Z_b)$ and $\prod_b M(Z_b)$ there is exactly one canonical map, the *oplax product-comparison*

$$\mathrm{st}_{(Z_b)}\colon\ M(\textstyle\prod_b Z_b) \longrightarrow \prod_b M(Z_b), \qquad w \longmapsto \big(M(\pi_b)\, w\big)_b,$$

built from M and the product projections alone. It exists for *every* functor M by the universal property of the target product — it is the structure map exhibiting M as an oplax monoidal functor for $(\mathsf{Set}, \times)$ — and no algebra structure on M is used to write it down.
The direction is forced. st runs $G_M T_M$-positions to $T_M G_M$-positions. Container morphisms run their position map *backward*, so st is the backward part of a morphism the *other* way,

$$\lambda_X\colon\ T_M G_M X \longrightarrow G_M T_M X, \qquad \text{forward } \mathrm{id}_{MS}, \quad \text{backward st},$$

a 2-cell $\lambda\colon T_M G_M \Rightarrow G_M T_M$: the standard orientation of a mixed distributive law of the comonad G_M over the monad T_M. The naive opposite orientation would need the reverse map $\prod_b M(Z_b) \to M(\prod_b Z_b)$, which is not canonical and, we show in §5, does not assemble into a law.

4.2. The entwining theorem.

Theorem 4.1 (The two feeds entwine). *Let M be a $\prod$-cointerpretation* Set-*monad, T_M, G_M its two container liftings, and $\lambda\colon T_M G_M \Rightarrow G_M T_M$ the 2-cell backward* st*. Then (T_M, G_M, λ) is an entwining structure on* Cont*: λ is natural and satisfies the four entwining axioms. Consequently G_M lifts to a comonad on* $\mathrm{Alg}(T_M)$ *and T_M lifts to a monad on* $\mathrm{Coalg}(G_M)$*, exhibiting a λ-bialgebra in the sense of Turi and Plotkin.*

Proof. Everything is identity on shapes, so each axiom localises to a position-level equation in Set, with backward maps composing contravariantly; localise at Z_b and write $\mathrm{st} = (M\pi_b)_b$.

- **Counit-G** (ε_G against λ). Backward, $M(\pi_b) \circ \eta_{\prod Z} = \eta_{Z_b} \circ \pi_b$: the naturality of η at π_b.

- **Unit-T.** At a single shape the product collapses to one factor; there st $= M(\mathrm{id}) = \mathrm{id}$ and the Mendler i is the singleton projection $= \mathrm{id}$, so both sides are id_{MZ}.
- **Comult-G** (δ_G against λ). Backward, $M(\pi_b)\circ\mu_{\prod Z} = \mu_{Z_b}\circ MM(\pi_b)$: the naturality of μ at π_b. Threading the two paths componentwise, both equal $(\mu_{Z_b}\circ MM(\pi_b))_b$.
- **Mult-T.** The only axiom touching the Mendler j. Both paths are built from st and the reindexing of a product along the leaf-map of μ^M; since st is componentwise M applied to a projection, it commutes with that reindexing ($M(\pi_{\rho(b,c)}) = M(\pi_{(b,c)})\circ M(\mathrm{reindex})$), and the two backward maps coincide. This is the naturality of st with respect to product-reindexing, i.e. the j-naturality diagram of Ahman–Bauer [1, Def. 6.2].

Naturality of λ in X is naturality of st and of the projections. The lifting to $\mathrm{Alg}(T_M)$ and $\mathrm{Coalg}(G_M)$ is the standard consequence of the entwining axioms [13]. □

Remark 4.2 (The one-line meaning of λ). The entwining says one thing: *M oplax-preserves products, coherently with its unit and multiplication, transported onto positions.* Fibrationally, $G_M = (M^{\mathrm{op}})_*$ is vertical and T_M covers the base monad M; λ is the Beck–Chevalley-type 2-cell comparing *apply M on the base, then fatten positions* against *fatten positions, then apply M on the base.* The comparison exists one way for free — that is st — and the four axioms are exactly its coherence with the two monad structures M carries on the two feeds.

Remark 4.3 (The mult-T index-chase). Axiom mult-T is the only one whose full symbolic verification depends on the specific weak Mendler j. The conceptual mechanism above — st commutes with the product-reindexing that is j — is complete; the fully spelt-out index-chase over an arbitrary j is mechanical and is not written out. It is machine-verified for the $\prod$-cointerpretation examples across three containers, including the *branching* commutative case $M = \mathsf{Pf}$ (union/restriction j), $M = \mathsf{Maybe}$, and $M = \mathsf{Writer}$ over both an abelian and a non-commutative log. The other three axioms are complete as stated.

The bialgebra face holds for *every* M. In particular nondeterminism (Pf), lists, and — for any $\prod$-cointerpretation presentation — probabilistic branching all carry the λ-bialgebra: the two feeds of a branching monad still entwine in the standard orientation, and G_M still acts on T_M-algebras. This is the sense in which the denotational unification of effects and coeffects on containers is unconditional. What branching costs is the *other* orientation, to which we now turn.

5. One direction only: the branching obstruction

The opposite orientation $G_M T_M \Rightarrow T_M G_M$ needs the *lax* comparison $\prod_b M(Z_b) \to M(\prod_b Z_b)$, which not every monad has; a commutative monad supplies the cartesian one (for Pf, $(S_b)_b \mapsto \prod_b S_b$, the set of choice-tuples). Even when it exists, it fails to be a distributive law the moment M branches.

Example 5.1 (Pf breaks the reverse orientation)**.** Take $M = \mathsf{Pf}$ and the container $X = (\{a,b\},\ a \mapsto \{0,1\},\ b \mapsto \{0\})$. The lax $G_M T_M \Rightarrow T_M G_M$ map satisfies unit-T, counit-G, and comult-G, but fails mult-T. At the double shape $\{\{a,b\},\{a\}\} \in \mathsf{Pf}\,\mathsf{Pf}\,S$ the two paths return, on one side, the *correlated* two-element set $\{((1,0),(1)),\ ((0,0),(0))\}$ — a union of products — and on the other the full four-element *product of unions*. The multiplication $\mu = \bigcup$ merges the leaf a shared by $\{a,b\}$ and $\{a\}$, and the cartesian lax map cannot see that the two choices at that shared leaf must agree: *union of products* $\neq$ *product of unions*.

The obstruction is **branching, not non-commutativity**. Pf is commutative and still fails; what breaks the law is μ identifying leaves, which needs a shape with at least two positions so that $\bigcup$ can overlap. When M has arity at most one no two leaves are ever shared, μ can identify none, mult-T holds trivially, and *both* orientations are entwinings at once. The dichotomy is clean:

M	$T_M G_M \Rightarrow G_M T_M$ (oplax st)	$G_M T_M \Rightarrow T_M G_M$ (lax)
arity ≤ 1 ($\mathsf{Maybe}, \mathsf{Writer}, \mathrm{Id}$)	entwining ✓	entwining ✓
branching ($\mathsf{Pf}, \mathsf{List}, \dots$)	entwining ✓ (**universal**)	fails mult-T ✗

So the interaction of the two feeds is real, and it lives in the standard orientation, where it holds for all M; the orientation one might guess first is the one branching forbids. The same variance that forced G_M to be a comonad also forces the single direction in which the monad feed can pass through it.

6. The arrow face: composing effect–coeffect programs

We now read the reverse orientation as a *composition rule*. An *effect–coeffect arrow* $p \rightsquigarrow q$ is a container morphism $G_M p \to T_M q$: it reads its input through the coeffect G_M and delivers its output through the effect T_M. This is the container analogue of a coKleisli-of-G-then-Kleisli-of-T program $GA \to TB$, and it is what one *sequentially composes.*

6.1. **The biKleisli category and its compositor.** The arrows $G_M p \to T_M q$ are the morphisms of the coKleisli category of the comonad G_M lifted to the Kleisli category $\mathrm{Kl}(T_M)$ — the biKleisli category of the pair [13, 15]. Explicitly, the identity $p \rightsquigarrow p$ is $\eta^T_p \circ \varepsilon_p$, and the composite of $f\colon G_M p \to T_M q$

and $g\colon G_M q \to T_M r$ is

$$G_M p \xrightarrow{\delta_p} G_M G_M p \xrightarrow{G_M f} G_M T_M q \xrightarrow{\kappa_q} T_M G_M q \xrightarrow{T_M g} T_M T_M r \xrightarrow{\mu^T_r} T_M r.$$

The middle step is a natural transformation $\kappa\colon G_M T_M \Rightarrow T_M G_M$: the composite *commutes the effect* T_M *out through the coeffect* G_M, from $G_M T_M q$ to $T_M G_M q$. This is the *reverse* of the proved λ of Theorem 4.1; on positions its backward map is the *lax* product-comparison $\prod_b M(Z_b) \to M(\prod_b Z_b)$ of §5. Composition therefore requires exactly the orientation that branching obstructs.

Theorem 6.1 (Existence of the arrow category)**.** *For a* $\prod$*-cointerpretation* Set*-monad* M *the following are equivalent.*

(1) *The effect–coeffect arrows* $G_M p \to T_M q$ *form a category with identity* $\eta^T \circ \varepsilon$ *and the biKleisli composition above.*
(2) *There is a natural* $\kappa\colon G_M T_M \Rightarrow T_M G_M$ *satisfying the four mixed-distributive-law axioms E1′–E4′.*
(3) M *is non-branching.*

The unit laws correspond to E1′, E3′; associativity corresponds to ***E2′*** *(the mult-*T *axiom), which is the sole axiom that branching destroys.*

Proof. (1) $\Leftrightarrow$ (2) is the mixed analogue of Beck's theorem [13]: a lifting of G_M to $\mathrm{Kl}(T_M)$ is the same datum as a κ obeying E1′–E4′, and the coKleisli category of the lifted comonad is (1), with the hom-sets $\mathsf{Cont}(G_M p, T_M q)$, identity $\eta^T \circ \varepsilon$, and the displayed composite; the unit laws unwind to E1′/E3′ and associativity to E2′, which is precisely the statement that the lifted comonad preserves Kleisli composition. (2) $\Leftrightarrow$ (3) is the dichotomy of §5: the lax κ satisfies E1′, E3′, E4′ for every M and satisfies E2′ iff M is non-branching, the branching failure being the union-of-products versus product-of-unions discrepancy of Example 5.1. ☐

The equivalence (1) $\Leftrightarrow$ (3) is what makes the result sharp: the existence of an effect–coeffect arrow calculus on containers is a property of the effect monad M, namely non-branching, and the obstruction is exactly the entwining axiom that branching destroys. The signature is telling. For $M = \mathsf{Pf}$ the unit laws still hold and *only* associativity breaks; an explicit non-associative triple $f, g, h\colon G_M(A_1) \to T_M(A_1)$ (with $A_1 = (\{a,b\}; a \mapsto 2, b \mapsto 1)$, all forward $a \mapsto \{a\}, b \mapsto \{a,b\}$) differs at shape b, position $(1,0)$, returning $\varnothing$ on one association and $\{0\}$ on the other.

6.2. **The Freyd/Hughes structure, and the strength obstruction.** A category is not yet an arrow: an arrow has, in addition, an identity-on-objects functor from a cartesian base and a first operator. We take the tensor to be the cartesian product $\times$ on Cont. For a pure morphism $\varphi\colon p \to q$ set $\mathrm{arr}(\varphi) = \eta^T_q \circ \varphi \circ \varepsilon_p\colon G_M p \to T_M q$.

Proposition 6.2 (The pure functor)**.** arr *is an identity-on-objects functor* $\mathsf{Cont} \to \mathsf{Arr}_M$*:* $\mathrm{arr}(\mathrm{id})$ *is the biKleisli identity, and* $\mathrm{arr}(\psi \circ \varphi) = \mathrm{arr}(\varphi) \ggg \mathrm{arr}(\psi)$*. Its image is the pure/central subcategory.*

Proof. Functoriality is the counit/unit calculation: expanding the composite and using $\varepsilon \circ \delta = \mathrm{id}$ together with the entwining triangles E1′ ($\kappa \circ G_M\eta^T = \eta^T G_M$) and E3′ ($T_M\varepsilon \circ \kappa = \varepsilon T_M$), both of which hold for *every* M (§5), collapses the middle $\eta^T \cdots \varepsilon$ to give $\eta^T \circ \psi \circ \varphi \circ \varepsilon = \mathrm{arr}(\psi \circ \varphi)$. □

The coeffect side is unconditionally well-behaved; the effect side is where branching bites again.

Lemma 6.3 (Coeffects are always costrong)**.** *For every* M *there is a natural costrength* $\sigma_{p,c} \colon G_M(p \times c) \to G_M p \times c$*, forward* id*, backward at* (v,t) *the map* $M(Qv) \sqcup Q_c t \to M(Qv \sqcup Q_c t)$ *that is* $M(\mathrm{inl})$ *on the coeffect summand and* $\eta^M \circ \mathrm{inr}$ *on the fresh summand.*

Proof. G_M-positions are a *single* M-structure $M(Ps)$, with no product over leaves, so the map is M applied to inl and the unit on the new c-factor; no branching arises. Naturality is functoriality of M and naturality of η^M. □

Lemma 6.4 (Effects are strong iff non-branching)**.** *The effect monad* T_M *admits a* natural *tensorial strength* $\tau_{q,c} \colon T_M q \times c \to T_M(q \times c)$ *for* $\times$ *if and only if* M *is non-branching.*

Proof. Backward, τ is the distributivity map $d_{m,t} \colon \prod_{b \in \mathrm{lv}m}(Q(v_b) \sqcup Q_c t) \to (\prod_b Q(v_b)) \sqcup Q_c t$.

(⇐) If every m has $|\mathrm{lv}m| \le 1$ then this is, at a nullary shape, the coprojection $1 \to 1 \sqcup Q_c t$, and at a unary shape the identity on $Q(v_0) \sqcup Q_c t$ (a length-1 product is its unique factor). No choice is made, so τ is natural and satisfies the strength unit and multiplication axioms.

(⇒) Suppose some m_0 has $n = |\mathrm{lv}m_0| \ge 2$. The family d must be natural in q and c, i.e. a natural transformation of the functors $L(A,C) = \prod_b(A_b \sqcup C)$ and $R(A,C) = (\prod_b A_b) \sqcup C$ of the variables $A_b := Q(v_b)$, $C := Q_c t$, and equivariant under the reindexings of $\mathrm{lv}m_0$ realised by M's functoriality. Specialise every $A_b = \varnothing$: then $L(\varnothing, C) = C^n$ and $R(\varnothing, C) = C$, so d restricts to a natural map $C^n \Rightarrow C$. Since $C^n = \mathrm{Hom}(n,-)$, Yoneda gives $\mathrm{Nat}(\mathrm{Hom}(n,-),\mathrm{Id}) \cong n$: the natural maps $C^n \to C$ are exactly the n leaf projections, so $d|_{A=\varnothing} = \pi_i$ for a *fixed* distinguished leaf i. That distinguished leaf is the contradiction. For *symmetric* branching (e.g. Pf), M's functoriality realises the transposition of two leaves of m_0 as an automorphism fixing the shape; equivariance forces $\pi_i = \pi_j$, impossible on C^n with $|C| \ge 2$. For *ordered* branching (e.g. List), a leaf-reindexing morphism breaks the fixed choice likewise. Hence no natural d, so no natural τ. □

Remark 6.5 (Two faces of branching)*.* A *total* strength does exist for branching M — the “priority” rule (all-inl to the product, else the leftmost inr value) is total and even satisfies the strength unit and multiplication axioms

— but it is *not natural*: it hard-codes the distinguished leaf that naturality forbids. So the strength obstruction is genuinely distinct from the associativity obstruction of Theorem 6.1: the latter is μ^T *merging* (E2′, union-of-products), the former is leaf-*symmetry* (naturality). Branching disables the arrow through two independent axioms, both equivalent to arity ≤ 1.

Theorem 6.6 (The arrow is Freyd iff non-branching)**.** *For non-branching M, with σ_G (Lemma 6.3) and τ_T (Lemma 6.4) and* $\mathrm{first}(f) = \tau_{q,c} \circ (f \times \mathrm{id}_c) \circ \sigma_{p,c}$*, the data* $(\mathsf{Arr}_M, \mathrm{arr}, \ggg, \mathrm{first})$ *satisfy all the Hughes arrow laws;* Arr_M *is a Hughes arrow, equivalently a Freyd category over the cartesian base* $(\mathsf{Cont}, \times)$*. For branching M,* Arr_M *is not even a category (Theorem 6.1), and* first *is not definable (T_M has no natural strength, Lemma 6.4). Hence*

$$\boxed{\mathsf{Arr}_M \textit{ is a Hughes arrow / Freyd category} \iff M \textit{ is non-branching.}}$$

Proof. The constructions arr, first are uniform in M; the abstract packaging is the standard theorem that the biKleisli category of a strong monad and a costrong comonad with a compatible distributive law is a Freyd/arrow category [15, 12, 2, 8]. The container-specific content — that σ_G, τ_T, κ satisfy the required (co)strength and compatibility coherences — holds for non-branching M by Lemmas 6.3–6.4 and Theorem 6.1, and is verified exhaustively for the representatives Maybe and $\mathsf{Writer}/\mathbb{Z}_2$, which span the normal form of §7. The branching direction is Theorem 6.1 and Lemma 6.4. ☐

Remark 6.7 (Arr_M is Atkey's biKleisli Arrow; branching is transverse to Atkey's index)*.* Atkey [2] shows that Hughes arrows are *not* the same thing as Freyd categories: an arrow is strictly more general — it admits a second, comonad-structured input that a Freyd category lacks — and the precise correspondence is *arrows* $\cong$ *closed **indexed** Freyd categories*, the extra generality controlled by an *index* that records the comonadic input. His running construction (§2, after Uustalu–Vene [15]) is exactly ours: from a comonad W, a monad T, and a distributive law $WT \Rightarrow TW$ he forms the biKleisli arrow $\mathrm{Ar}(x, y) = [Wx \to Ty]$; with $W = G_M$, $T = T_M$, and $\lambda = \kappa$ this is *verbatim* our $\mathsf{Arr}_M(p, q) = \mathsf{Cont}(G_M p, T_M q)$.

It is tempting to read Theorem 6.6 as an *index-collapse* — non-branching M trivialising Atkey's index so that Arr_M becomes a plain (non-indexed) Freyd category — but that reading is *false*, and the way it fails is informative. Atkey's index records the *coeffect* input $W = G_M$, and G_M is trivial exactly when M is: $G_M = \mathrm{Id}$ iff $M = \mathrm{Id}$, immediate from $G_M(S, P) = (S, M \circ P)$. Index-collapse is therefore the single condition $M = \mathrm{Id}$, *strictly stronger* than non-branching: Maybe and Writer are non-branching with $G_M \neq \mathrm{Id}$, and Theorem 6.6 still makes their Arr_M a genuine Freyd category — through the *strength* of T_M, not through any collapse of the coeffect index. The branching axis and Atkey's index axis are thus *orthogonal*: index-collapse ($M = \mathrm{Id}$) is a proper sub-condition of non-branching, and Atkey's index measures a coeffect that persists across the entire non-branching class. *Non-branching is not index-collapse.*

What branching controls is whether the biKleisli construction lands in Atkey's landscape at all: for branching M the compositor κ is not even a distributive law (E2$'$ fails, Theorem 6.1), so Arr_M degrades below an indexed Freyd category — below even an arrow. The branching obstruction sits on the *arity* axis, transverse to Atkey's monad $\subset$ Freyd $\subset$ indexed-Freyd $\subset$ arrow stratification of the coeffect; whether the resulting (Boolean) arity dichotomy admits any graded refinement is taken up in §10.

Remark 6.8 (The pure and effectful sub-calculi). The image of arr is the pure/central subcategory. The *effect* sub-calculus is the Kleisli category $\mathrm{Kl}(T_M)$ (collapse G_M by ε): container programs with the Ahman–Bauer effect. The *coeffect* sub-calculus is the coKleisli category $\mathrm{coKl}(G_M)$ (collapse T_M by η^T): container programs reading the transfer-comonad context. The arrow Arr_M *fuses* them, and Theorem 6.1 says the fusion is coherent exactly when M does not branch.

7. The positive class: writer with absorbing exceptions

Theorems 6.1–6.6 leave a name to be attached: which monads are the non-branching ones? We classify them completely.

Theorem 7.1 (Classification)**.** *For a cartesian $\prod$-cointerpretation* Set*-monad M the following are equivalent:*

1. *the arrow category Arr_M exists;*
2. *M is non-branching;*
3. *$M \cong E + A \times (-)$ as a functor, for some sets E, A;*
4. *M is a* writer-with-absorbing-exceptions *monad: $MX = E + A \times X$ with $(A, \cdot, e)$ a monoid, $(E, \odot)$ a left A-set, writer unit $\eta_X(x) = (e, x)$, and*

$$\mu_X\colon\ \mathrm{inl}\, e' \mapsto \mathrm{inl}\, e', \quad \mathrm{inr}(a, \mathrm{inl}\, e') \mapsto \mathrm{inl}(a \odot e'), \quad \mathrm{inr}(a, \mathrm{inr}(a', x)) \mapsto \mathrm{inr}(a \cdot a', x).$$

Moreover, for every such M and every container p the compositor κ satisfies all four axioms E1$'$–E4$'$; in particular the associativity axiom E2$'$ holds throughout the class, so Arr_M is a category for the entire class, not only for Maybe *and* Writer*.*

Proof. (1) $\Leftrightarrow$ (2) is Theorem 6.1.

(2) $\Leftrightarrow$ (3): a polynomial functor is $MX = \sum_{s \in M1} X^{\mathrm{ar}(s)}$. If $|\mathrm{ar}(s)| \le 1$ for all s, partition the shapes as $E = \{s : \mathrm{ar}(s) = \varnothing\}$ and $A = \{s : |\mathrm{ar}(s)| = 1\}$; then $X^{\mathrm{ar}(s)}$ is 1 on E and X on A, so $MX \cong E + A \times X$ naturally. Conversely $E + A \times (-)$ has all arities ≤ 1.

(3) $\Rightarrow$ (4): let $MX = E + A \times X$ carry a monad. By naturality (Yoneda for the identity domain), the unit η is either a constant $x \mapsto \mathrm{inl}\, e_0$ or $x \mapsto \mathrm{inr}(a_0, x)$; the constant form violates the left-unit law, so $\eta_X(x) = \mathrm{inr}(e_0, x)$ for a distinguished $e_0 \in A$ (the writer unit). Naturality similarly forces μ to the schematic form above, carrying the variable x exactly when the outer and inner A-labels multiply into A. Writing $N := E \sqcup A$ with the induced binary

operation, the unit laws make e_0 a two-sided unit in A, and the associativity of μ unwinds to the associativity of N; E is automatically a two-sided ideal of left zeros. In the *cartesian* case μ^M preserves the leaf-count, which rules out *aborting* ($a \cdot a' \in E$ for $a, a' \in A$ would destroy the single leaf), so A is a submonoid and σ restricts to a unital, associative action $\odot\colon A \times E \to E$: E is a left A-set, statement (4). (4) $\Rightarrow$ (3) is immediate.

Finally, the last clause. Non-branching means every $P^\star$-product occurring anywhere is a product over ≤ 1 factor, so κ's backward map is, shape by shape, either the identity (one leaf: rewrap the single factor) or the monad unit $\eta^M\colon 1 \to M1$ (no leaf: the empty product). Reading each axiom E1′–E4′ as an equality of Cont-morphisms: on shapes all four reduce to the monad/comonad shape-laws of M; on positions the only nontrivial content sits at unary shapes, where $\kappa = \mathrm{id}$ and the axiom collapses to the corresponding structural law of M on the single leaf — for E2′, the associativity of $\cdot$ in A, which holds. The ≥ 2-leaf comparison that breaks E2′ for branching M never forms. □

Remark 7.2 (The Set-monad level, and why aborting is excluded). One level up, the functors $E + A \times (-)$ that carry *any* monad structure are exactly the monoids $N = E \sqcup A$ with unit in A and E a two-sided ideal of left zeros — the multiplication of two unary shapes may *abort* into E (e.g. the nilpotent $\{e, z, 0\}$ with $z^2 = 0 \in E$, giving $MX = 1 + 2 \times X$). These aborting monads are genuine Set-monads but are *not cartesian*: their μ destroys a leaf, so Ahman–Bauer's T_M is undefined for them and they fall outside the arrow story. Within the class where T_M exists, the non-branching monads are exactly the non-aborting ones — the writer-with-absorbing-exceptions monads. (With trivial action $\odot$, $E + A \times X$ is the writer transformer over the exception monad, $\mathsf{WriterT}_A(\mathsf{Exc}_E)(X)$; general $\odot$ twists the discard by a left A-action on E.)

Remark 7.3 (Scope). Both λ and κ, and the strength τ, use the product structure of $P^\star$; whether a canonical law exists for a general (non-$\prod$) weak Mendler algebra is open, and out of scope. The Freyd base is the cartesian $\times$; a monoidal (non-cartesian) arrow over the Dirichlet tensor $\otimes$ would need a different strength story and is a separate question. No finite $\prod$-cointerpretation monad is simultaneously branching and non-commutative (List is the witness, but infinite), so that corner of the entwining theorem is proved by naturality and untested computationally.

7.1. **Related conditions: non-branching is genuinely new.** The dividing line of Theorems 6.1–7.1 is *non-branching*. One should check that this is not a disguised form of one of the two classical properties of a Set-monad — Kock *commutativity* [10, 11] and Kock/Jacobs *affineness* $M1 \cong 1$ [7] (the sense of Remark 2.1, not the polynomial-degree sense we deliberately avoid). It is not: writing

$$\text{P1} = \text{non-branching}, \qquad \text{P2} = \text{commutative}, \qquad \text{P3} = \text{affine } (M1 \cong 1),$$

the three are pairwise logically independent, with a single implication among them pointing the other way (P1∧P3⇒P2). The commutativity of the members of the positive class is itself sharp:

Lemma 7.4 (Commutativity criterion for the class)**.** *A writer-with-absorbing-exceptions monad $MX = E + A \times X$ (Theorem 7.1) is commutative if and only if* (i) *A is a commutative monoid,* (ii) $|E| \leq 1$*, and* (iii) *the action $\odot$ is trivial. In particular* Maybe *($|E| = 1$, $A = 1$) and* Writer_A *with A abelian are commutative, whereas* Writer_{N_3} *over a non-commutative monoid N_3 and the exception monad $E + (-)$ with $|E| \geq 2$ are not.*

Proof. M is commutative iff Kock's two canonical maps $MX \times MY \to M(X \times Y)$ agree. Evaluating both on the four kinds of pair in $(E + A \times X) \times (E + A \times Y)$: on $(\mathrm{inl}\, e, \mathrm{inl}\, e')$ they return $\mathrm{inl}\, e$ and $\mathrm{inl}\, e'$ (equal for all inputs iff $|E| \leq 1$); on $(\mathrm{inl}\, e, \mathrm{inr}(b, y))$ they return $\mathrm{inl}\, e$ and $\mathrm{inl}(b \odot e)$, and symmetrically on the mirror pair (equal iff $\odot$ is trivial); on $(\mathrm{inr}(a, x), \mathrm{inr}(b, y))$ they return $\mathrm{inr}(a \cdot b, (x, y))$ and $\mathrm{inr}(b \cdot a, (x, y))$ (equal iff A is commutative). All four agree iff (i)–(iii). □

Proposition 7.5 (Non-branching is independent of commutativity and affineness)**.** *Each of the three 2×2 faces of the cube* (P1, P2, P3) *is fully realised by explicit* Set*-monads; hence no one of the three properties implies or precludes another. Concretely:*

(P1, P2, P3)	*witness M*	*values*
(T, T, T)	Id	*ar* ≤ 1, *comm*, $M1 = 1$
(T, T, F)	$\mathsf{Maybe} = 1 + (-)$	*ar* ≤ 1, *comm*, $M1 = 2$
(T, F, T)	— *impossible* —	*Remark 7.6*
(T, F, F)	$\mathsf{Writer}_{N_3} = N_3 \times (-)$	*ar* ≤ 1, *non-comm*, $M1 = 3$
(F, T, T)	$\mathcal{P}^+$ (*non-empty powerset*)	*branch*, *comm*, $M1 = 1$
(F, T, F)	Pf	*branch*, *comm*, $M1 = 2$
(F, F, T)	*free idempotent magma*	*branch*, *non-comm*, $M1 = 1$
(F, F, F)	*free magma (binary trees)*	*branch*, *non-comm*, $M1 = \infty$

Here $N_3 = \{1, a, b\}$ is the three-element non-commutative monoid with $a \cdot x = a$ and $b \cdot x = b$ for $x \in \{a, b\}$.

Proof. Each 2×2 face is a projection of the table (e.g. P1×P2 is realised by Id, Writer_{N_3}, Pf, and the free magma), so every pair of properties attains all four truth-values and pairwise independence follows. The property values are established as: **P1** by the support certificate of §2.2 (the $E + A \times (-)$ witnesses have all supports ≤ 1; $\mathcal{P}^+$, Pf and both magmas carry a support-2 element); **P2** by Lemma 7.4 for the non-branching witnesses, by the classical commutativity of the semilattice monads $\mathcal{P}^+$, Pf [11], and by the failure of the medial/interchange law $(a * b) * (c * d) = (a * c) * (b * d)$ — forced by Kock's theorem that a commutative monad has homomorphic operations —

in explicit small models of the two magmas; **P3** by the displayed values of $|M1|$. Full witnesses, the four-row commutativity chase, and an exhaustive machine sweep are in the author's proof notes. □

Remark 7.6 (The one dependency). The three properties are *not* jointly independent: the cube has exactly one empty cell, $(\mathrm{P1}, \neg\mathrm{P2}, \mathrm{P3})$, because **non-branching and affine imply commutative**. If $M \cong E + A \times (-)$ is non-branching then $M1 \cong E + A$, so affineness $|M1| = 1$ forces $(|E|, |A|) \in \{(0, 1), (1, 0)\}$, i.e. $M \cong \mathrm{Id}$ or the constant monad 1, both commutative. Equivalently, *every non-commutative affine monad is branching*: affineness collapses the non-branching world to the two trivial monads and so cannot express non-branching. This lone implication — pointing away from non-branching — is the only logical relation among P1, P2, P3, which confirms that "non-branching" is a genuinely new dividing line rather than a restatement of a classical monad condition.

8. A machine-checked instance

The non-branching monad $M = \mathsf{Maybe}$ furnishes the smallest genuinely effectful arrow category, and it has been formalised in Lean 4. The development `BiKleisliMaybe.lean` constructs the effect monad $T_{\mathsf{Maybe}}(S, P) = (1 + S,\ P^\star)$ as an instance of an abstract `MixedDistrib` structure — monad T, comonad G, compositor κ, and the four axioms E1′–E4′ including the associativity axiom E2′ — and derives the biKleisli composition together with its associativity law `acomp_assoc` from those axioms, with no `sorry`. This is, to the author's knowledge, the first machine-checked *associative* effect–coeffect arrow category on containers for a concrete non-branching monad. It verifies the load-bearing content of Theorem 6.1 in the smallest non-trivial case; the general E2′ across the whole class (Theorem 7.1) is proved in coordinates but not yet Lean-certified, and remains a natural next target.

9. Related work

The two feeds, and their sources. The effect monad T_M is Ahman–Bauer [1, Thm. 6.3]; the coeffect comonad G_M is the transfer (the author's, proved and Lean-verified). Neither paper forms the composites $T_M G_M$ / $G_M T_M$ or a distributive law between the feeds. The entwining and arrow formalism is classical: Beck [4]; Power–Watanabe [13]; Brzeziński–Majid [5]; Uustalu–Vene [15]; Hughes [6]; Power–Robinson [12]; Atkey [2]; Jacobs–Heunen–Hasuo [8]; Turi–Plotkin [14]. We deploy it where the base monad sits on positions and branching is the decisive parameter.
Distributive laws of container (co)monads (Purdy–Damato). Purdy and Damato [3] characterise distributive laws between container-monads and container-comonads *on* Set, including mixed monadic-over-directed laws, by an explicit data presentation. By contrast our T_M and G_M are a monad and a comonad on the category Cont itself, obtained from a single Set-monad M

by transferring it to shapes and to positions respectively; their (co)monads are container functors on Set, whereas ours are endofunctors of Cont, one level up. Consequently our mixed law λ and its arrow-orientation κ live in $[\mathsf{Cont}, \mathsf{Cont}]$ and are not instances of their classification. (Their Example 22 links monadic-container laws to matching pairs of monoid actions in the sense of Zappa–Szép, adjacent to the directed mode discussed below but at the level of monoids rather than of the entwining.)

Interaction laws (Katsumata–Rivas–Uustalu). The nearest neighbour by intent is the theory of *interaction laws of monads and comonads* [9], which answers effect/coeffect interaction by a *pairing* $\psi\colon TX \times DY \to X \times Y$, realised as a monoid object in a Chu construction over $([\mathcal{C}, \mathcal{C}], \mathrm{Day})$. That is emphatically *not* a distributive law — distributive laws are *inputs* to their machinery, never outputs — so the two accounts are orthogonal in kind, a pairing versus a compositor. What they share is only the *location* of the obstruction: their degeneracy theorems locate the same branching wall from the side of extensive categories that our arity criterion locates from the side of polynomial functors — convergent evidence that branching is the universal obstruction to composing effects with coeffects.

Neighbours read only in outline. Two further lines of work are close in intent and deserve a detailed comparison we defer here to a later reading. Modern higher-order *bialgebraic denotational semantics* (Goncharov, Milius, Tsampas, Urbat, and collaborators, ICFP 2026) builds bialgebras from a syntax Σ-algebra and a behaviour B-coalgebra glued by higher-order GSOS laws via locally final coalgebras — a different object from our monad-over-comonad mixed distributive law, in which both T_M and G_M arise as the two feeds of one Set-monad rather than as separate syntax and behaviour functors. Separately, Dumas, Duval and Reynaud (*Patterns for computational effects arising from a monad or a comonad*, 2013) pair a monad with its associated comonad as two dual but *separate* equational proof-patterns (Kleisli versus coKleisli), whereas we entwine an independent shapes-monad and positions-comonad via a genuine distributive law. We flag both as neighbours rather than cite them as sources, having so far consulted them only in outline.

Three modes of composition. The effect–coeffect arrow is one of three container-native modes of welding compositional systems the author has studied, and it is the branching-obstructed one. The other two are the Zappa–Szép product of two directed containers (obstructed by a cohomology class $[\omega] \in H^2$) and the state-graded category of stateful "Workers" (unobstructed, the grade accumulating). The three obstructions — a cohomology class, nothing, and a Boolean arity condition — are three genuinely different kinds of mathematics for three ways of gluing; they do not unify into a master theorem, and the honest synthesis is that *whether* two systems compose is, in each mode, a computable invariant of the parts, though which invariant depends on the axis.

10. Conclusion

One monad M on Set feeds a container along two axes: forward on shapes it is the effect monad T_M, backward on positions it is the coeffect comonad G_M. We have shown that these two feeds always entwine — the oplax product-comparison of M, read on positions, is a mixed distributive law $\lambda\colon T_M G_M \Rightarrow G_M T_M$ for every M — giving a Turi–Plotkin λ-bialgebra in which G_M acts on T_M-algebras and T_M on G_M-coalgebras, nondeterminism and all. The reverse compositor κ that would let one *sequentially compose* effect–coeffect programs $G_M p \to T_M q$ into a Hughes arrow / Freyd category exists if and only if M is non-branching, and that class is exactly the writer-with-absorbing-exceptions monads $E + A \times (-)$. The same monad, the same two feeds: the unification of effects and coeffects is unconditional as a bialgebra and conditional as an arrow, and branching is precisely what separates the two — obstructing the orientation of the interaction, not the interaction itself.

Several questions remain open and are, to us, the natural continuations.

- *Beyond the $\prod$-cointerpretation.* Both laws use the product structure of $P^\star$. Is there a canonical λ for a general (non-$\prod$) weak Mendler algebra? There is no evident map $M(P^\star) \to (MP)^\star$ without the product, so we suspect not, but a proof either way is missing (Remark 7.3).
- *A graded refinement?* Read through Atkey's arrows $\cong$ indexed Freyd categories (Remark 6.7), one might hope the Boolean dichotomy "Arr_M is a category iff arity ≤ 1" is the bottom rung of a *graded* tower indexed by arity. A companion analysis rules out the two natural gradings. Along the effect axis the arity invariant is already two-valued — a cartesian Set-monad has maximal arity in $\{\leq 1, \infty\}$, with no finite rung between, since an arity-n shape substituted into itself has arity n^2 — so "arity $\leq n$" names no monads for $2 \leq n < \infty$; and the leaf-count grade on arrows is destroyed by the very merging in μ^{T_M} that branching forces. A grade that survives, if one exists, must live on the *coeffect*: Atkey's index measures G_M, nontrivial for every $M \neq \mathrm{Id}$ and independent of branching, and whether a graded comonad on G_M organises Arr_M along that axis is left open.
- *The Dirichlet arrow.* The Freyd reading fixes the cartesian $\times$. Does the Dirichlet tensor $\otimes$ on Cont carry a *monoidal* (non-cartesian) effect–coeffect arrow, with a different strength story? The costrength of G_M is the easy half; the effect strength for $\otimes$ is the open one.
- *Full formalisation.* The Maybe arrow category is machine-checked (§8); the associativity axiom E2′ across the whole writer-with-exceptions class (Theorem 7.1) is proved in coordinates but not yet in Lean. Certifying it, and the general mult-T index-chase for λ (Remark 4.3), would close the two mechanical gaps.

References

[1] D. Ahman, A. Bauer. *Comodule representations of second-order functionals.* arXiv:2409.17664; J. Log. Algebraic Methods Program. **146** (2025).

[2] R. Atkey. *What is a categorical model of arrows?* Electronic Notes in Theoretical Computer Science **229**(5) (2011) 19–37.

[3] C. Purdy, D. Damato. *Distributive laws of monadic containers.* arXiv:2503.17191; Proc. CALCO 2025.

[4] J. Beck. *Distributive laws.* In: Seminar on Triples and Categorical Homology Theory, Lecture Notes in Mathematics **80**, Springer (1969) 119–140.

[5] T. Brzeziński, S. Majid. *Coalgebra bundles.* Communications in Mathematical Physics **191** (1998) 467–492.

[6] J. Hughes. *Generalising monads to arrows.* Science of Computer Programming **37**(1–3) (2000) 67–111.

[7] B. Jacobs. *Affine monads and side-effect-freeness.* Proc. CMCS 2016, LNCS **9608**, Springer (2016) 53–72.

[8] B. Jacobs, C. Heunen, I. Hasuo. *Categorical semantics for arrows.* Journal of Functional Programming **19**(3–4) (2009) 403–438.

[9] S. Katsumata, E. Rivas, T. Uustalu. *Interaction laws of monads and comonads.* Proc. FoSSaCS 2020, LNCS **12077**, 356–375; arXiv:1912.13477.

[10] A. Kock. *Monads on symmetric monoidal closed categories.* Archiv der Mathematik **21** (1970) 1–10.

[11] A. Kock. *Closed categories generated by commutative monads.* Journal of the Australian Mathematical Society **12**(4) (1971) 405–424.

[12] J. Power, E. Robinson. *Premonoidal categories and notions of computation.* Mathematical Structures in Computer Science **7**(5) (1997) 453–468.

[13] J. Power, H. Watanabe. *Combining a monad and a comonad.* Theoretical Computer Science **280** (2002) 137–162.

[14] D. Turi, G. Plotkin. *Towards a mathematical operational semantics.* Proc. LICS 1997, IEEE (1997) 280–291.

[15] T. Uustalu, V. Vene. *Comonadic notions of computation.* Electronic Notes in Theoretical Computer Science **203**(5) (2008) 263–284.

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